algorithm as possible—the architecture, the hyperparameters used, and sometimes the random seed we used. However, we hope we've developed an algorithm that is robust and that the particular set of random numbers generated doesn't matter to the performance of the algorithm.

6.5.5. Scaling linearly

Because the OpenAI researchers reduced the volume of data sent between the nodes, adding nodes did not affect the network significantly. They were able to scale to over a thousand workers linearly.

Scaling linearly means that for every machine added, we receive roughly the same performance boost as we did by adding the previous machine. This is denoted by a straight line on a graph of performance over resources, as seen in figure 6.16.

Figure 6.16. Figure recreated from the OpenAI "Evolutionary Strategies as a Scalable Alternative to Reinforcement Learning" paper. The figure demonstrates that as more computing resources were added, the time improvement remained constant.

